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B.Tech. Degree VII Semester Supplementary Examination May 2024

ME 19-205-0702 VIBRATION AND NOISE CONTROL

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Model a given physical system using spring, mass and damping elements.

CO2: Formulate the governing differential equations of single dof, multi dof and continuous systems.

CO3: Solve the governing equations and interpret the results.

CO4: Understand the effects of noise and the control measures adopted.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

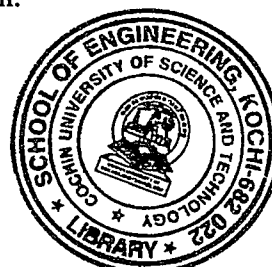
		(8 × 3 = 24)	Marks	BL	CO	PO
I.	(a)	A body of 5 kg is supported on a spring of stiffness 200 N/m and has dashpot connected to it which produces a resistance of 0.002 N at a velocity of 1 cm/s. In what ratio will the amplitude of vibration be reduced after 5 cycles.	3	L3	1	1
	(b)	What do you understand by support excited motion?	3	L3	1	2
	(c)	Explain the working principle of accelerometer.	3	L1	4	1
	(d)	Using Lagrange's equation, determine the equation of motion of a two degrees of freedom vibrating system with a neat sketch.	3	L2	4	2
	(e)	Apply Rayleigh's method to find the natural frequency of transverse vibration of shaft carrying two point loads.	3	L3	3	1
	(f)	Explain the concept of 'Whirling speed of shaft'.	3	L1	4	2
	(g)	Define continuous and intermittent noise.	3	L1	4	3
	(h)	Write a short note on noise exposure limits.	3	L2	4	2

PART B

(4 × 12 = 48)

II.	(a)	A gun barrel of mass 600 kg has a recoil spring of stiffness 294 kN/m. If the barrel recoils 1.3 m on firing determine	8	L3	1	2
	(i)	the initial recoil velocity of the barrel.				
	(ii)	the critical damping co-efficient of the dash pot which is engaged at the end of the recoil stroke.				
	(iii)	the equation of motion of the gun barrel if the time of recoil is of time period.				
	(b)	A torsional pendulum when immersed in oil indicates its natural frequency of 200 Hz. But when it was put to vibration in vacuum having no damping, its natural frequency was observed as 250 Hz. Find the value of damping factor of the oil.	4	L3	1	2

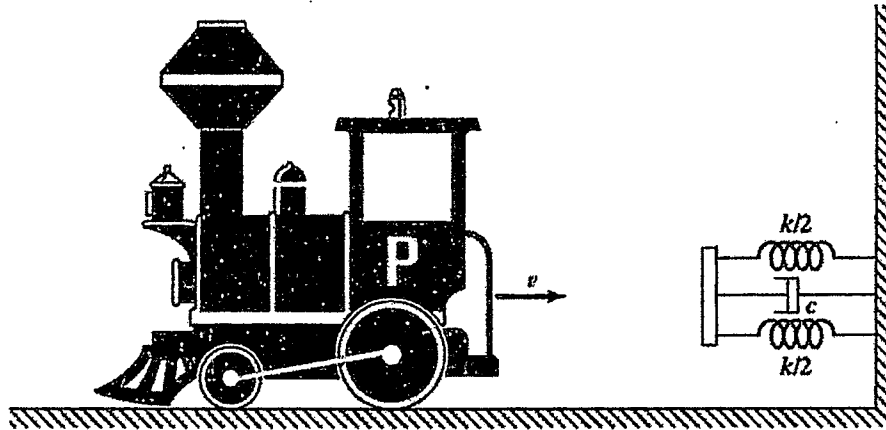
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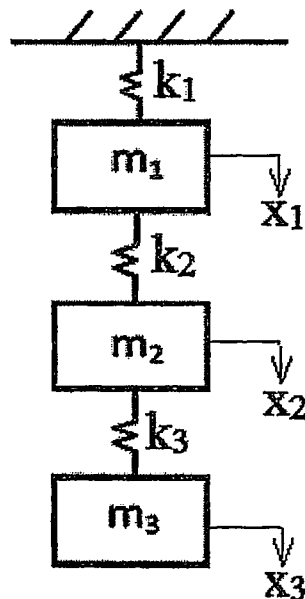
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- | | Marks | BL | CO | PO |
|--|-------|----|----|----|
| III. (a) The mass of an electric motor is 125 kg and it runs at 1500 r.p.m. The armature mass is 35 kg and its C.G. lies 0.05 cm from the axis of rotation. The motor is mounted on five springs of negligible damping so that the force transmitted is one-eleventh of the impressed force. Assume that the mass of the motor is equally distributed among the five springs. Determine
(i) stiffness of each spring.
(ii) dynamic force transmitted to the base at the operating speed.
(iii) natural frequency of the system. | 6 | L3 | 1 | 2 |
| (b) A railroad car of mass 2,000 kg traveling at a velocity of $v = 10$ m/s is stopped at the end of the tracks by a spring-damper system, as shown in figure. If the stiffness of the spring is $k = 80$ N/mm and the damping constant is $c = 20$ Ns/mm, determine
(i) the maximum displacement of the car after engaging the springs and damper.
(ii) the time taken to reach the maximum displacement. | 6 | L3 | 1 | 2 |



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|---|----|----|---|---|
| IV. Determine the natural frequencies and mode shapes of the system shown in the figure. Assume $m_1 = m_2 = m_3 = m$ and $k_1 = k_2 = k_3 = k$. | 12 | L3 | 2 | 2 |
|---|----|----|---|---|



OR

(Continued)

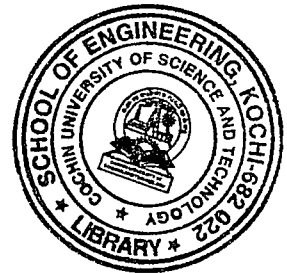
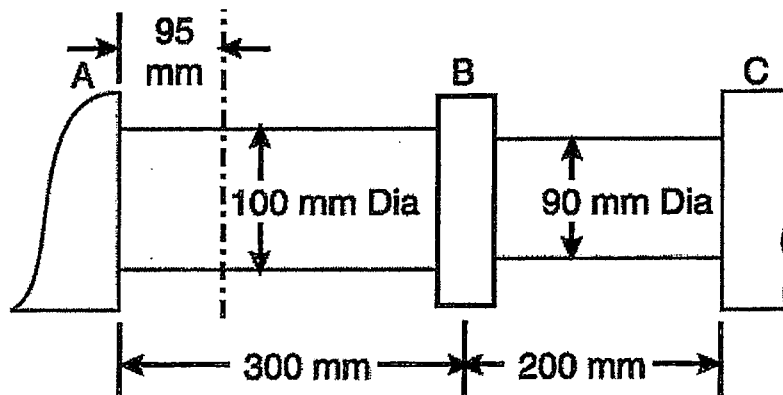
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Marks	BL	CO	PO
12	L3	2	2

- V. A motor generator set, as shown in figure, consists of two armatures A and C connected with a flywheel between them at B. The modulus of rigidity of the connecting shaft is 84 GPa. The system can vibrate torsionally with one node at 95 mm from A, the flywheel being at antinode. Find

- the position of another node.
- the natural frequency of vibration.
- the radius of gyration of the armature C.

Particulars	A	B	C
Radius of gyration, in mm	300	375	---
Mass in kg	400	500	300



- VI. A shaft 1.5 m long, supported in short flexible bearings at the ends carries two wheels each of 50 kg mass. One wheel is situated at the centre of the shaft and the other at a distance of 375 mm from the centre towards left. The shaft is hollow of external diameter 75 mm and internal diameter 40 mm. The density of the shaft material is 7700 kg/m^3 and its modulus of elasticity is 200 GN/m^2 . Find the lowest whirling speed of the shaft, taking into account the mass of the shaft.

OR

- VII. Derive the equation for transverse vibration of beams with solutions.
- VIII. (a) Explain the following terms
- Sound Pressure Level.
 - Sound Intensity Level.
 - Loudness.
 - Octave Bands.
- (b) Six machines operating individually, makes sound pressure levels of 78 dB, 81 dB, 81 dB, 79 dB, 72 dB and 65 dB respectively. Determine the overall sound pressure level when all of them operate simultaneously.

OR

- IX. (a) What are the various effects of continuous exposure of noise on human body?
- (b) Explain the various methods used in controlling industrial noise.

Bloom's Taxonomy Levels

L1 - 15.8%, L2 - 21.1%, L3 - 63.1%.
